



Real-time Pedestrian Detection and Safety Alert System for Signalized Intersections using Edge AI

18-786: Introduction to Deep Learning

Anirud Durani, Abhishek Reddy, Jonathan Ali

Motivation

- Pedestrian injuries at intersections are a critical urban challenge.
- Centralized traffic systems suffer latency and scalability limitations.
- Goal: Real-time pedestrian detection + alerting on Jetson Nano.

System Pipeline Overview

	Functional Component <i>(What it Does)</i>	Implementation Approach <i>(How it's Done)</i>
1.	Real-time Detection & Tracking	YOLO + DeepSORT
2.	Behavior Intent Classification	YOLO + LSTM
3.	Trajectory & Intent Prediction	LSTM Motion Forecasting
4.	Edge Deployment	Jetson Nano + DeepStream SDK

Datasets

1. Live Video Source
 - a. Location: Toomer's Corner, City of Auburn
 - b. Webcam Stream: YouTube Live Feed
(https://www.youtube.com/live/5WN2PJ_Qxjs?si=5YiRTKo3yZPRcxXc)

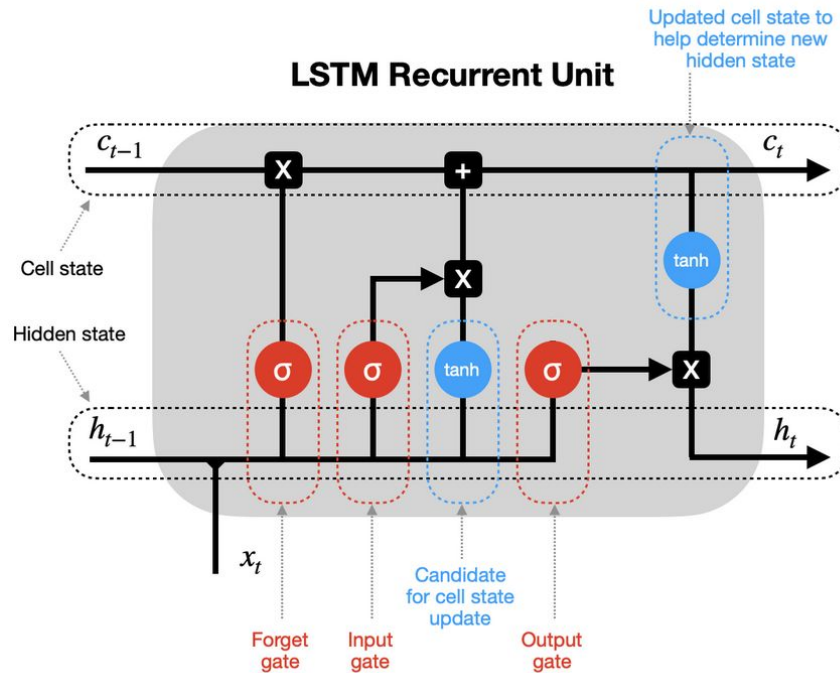


Dataset Annotations

Annotated Object Classes

- Vehicles
 - Car
- Traffic Signals
 - Traffic Light (Red)
 - Traffic Light (Yellow)
 - Traffic Light (Green)
- Pedestrian Intentions
 - Crossing on Crosswalk (pedestrian_cross_walk)
 - Standing at Crosswalk (pedestrian_cross_stand)
 - Standing without Intention to Cross (pedestrian_nocross_stand)
 - Walking outside Crosswalk without Crossing Intent (pedestrian_nocross_walk)

Long-Short Term Memory (LSTMs)



Data & Parameters

frame	track_id	class_id	class_name	co x1	y1	x2	y2	
5	1	2	car		204	650	350	753
5	2	3	traffic_light_green		131	428	228	542
5	3	1	traffic_light_red		1322	350	1430	490
5	4	3	traffic_light_green		415	409	485	516
5	5	7	ped_nocross_stand		1301	644	1347	732
5	6	1	traffic_light_red		1664	353	1748	509
5	7	2	car		1746	685	1918	876

- Object Filtering
- Feature Engineering
- Normalization
- Input/Output Construction

Train	Test
464 Unique IDs 99058 sequences	117 Unique IDs 25551 sequences
103690 total frames	26721 total frames

Pedestrian Intent LSTM

Architecture:

- Input Size: 3 (x, y, intent label at t)
- Sequence Length: 10
- LSTM Layers: 1
- Hidden Units: 64
- Output Size: 2 or 4 (intent classes)
- Final Layer: Linear(hidden → num_classes)

Hyperparameters:

- Loss Function: CrossEntropyLoss
- Optimizer: Adam
- Learning Rate: 1e-3
- Epochs: 20
- Batch Size: 32

Input Shape: (batch_size, 10, 3)

Trajectory Prediction LSTM

Architecture:

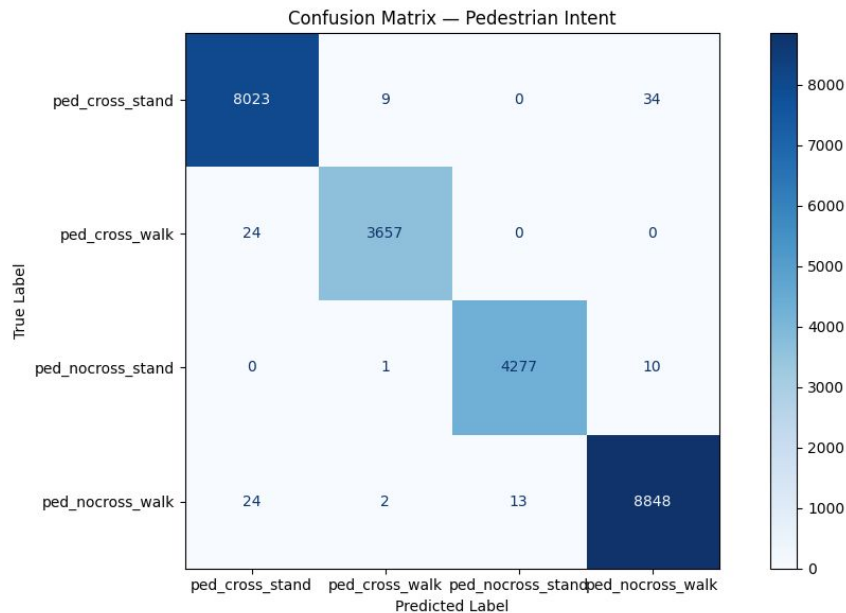
- Input Size: 3 (x, y, intent label at t)
- Sequence Length: 10
- LSTM Layers: 1
- Hidden Units: 64
- Output Size: 2 (x, y) coordinates
- Final Layer: Linear(hidden → 2)

Hyperparameters:

- Loss Function: MSELoss
- Optimizer: Adam
- Learning Rate: 1e-3
- Epochs: 20
- Batch Size: 32

Input Shape: (batch_size, 10, 3)

Predicting Pedestrian Intent via LSTMs



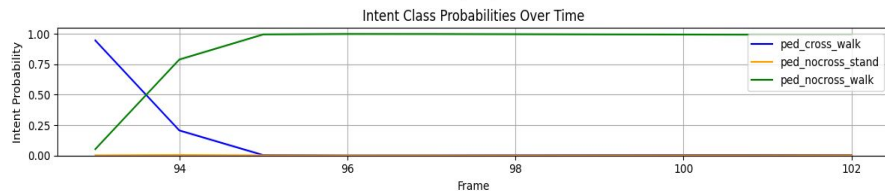
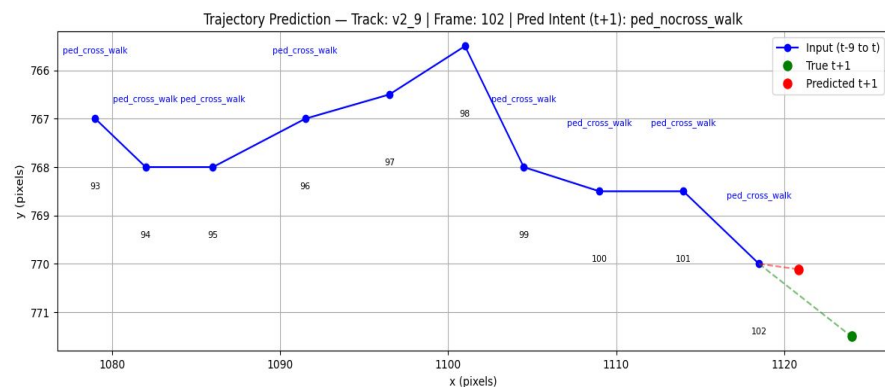
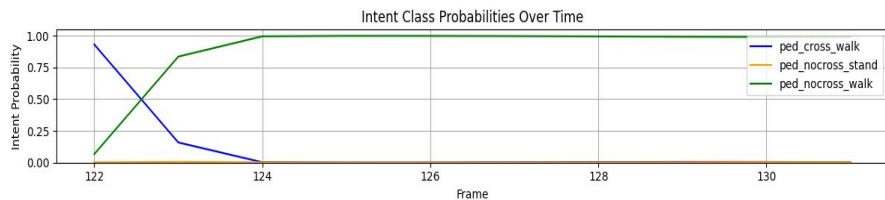
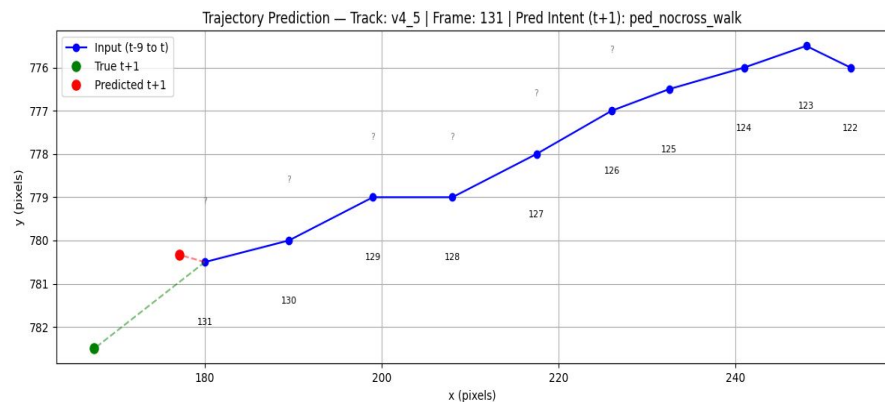
Classification Report:

	precision	recall	f1-score	support
ped_cross_stand	0.99	0.99	0.99	8066
ped_cross_walk	1.00	0.99	1.00	3681
ped_nocross_stand	1.00	1.00	1.00	4288
ped_nocross_walk	1.00	1.00	1.00	8887
accuracy			1.00	24922
macro avg	1.00	1.00	1.00	24922
weighted avg	1.00	1.00	1.00	24922

Predicting Pedestrian Trajectory via LSTMs

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Trajectory Model Evaluation:  
Mean Absolute Error (X):      1.18 px  
Mean Absolute Error (Y):      0.62 px  
Mean Absolute Error (Overall): 0.90 px  
Root Mean Squared Error:      1.85 px  
Final displacement error (FDE = Euclidean dist between predicted & true t+1): 1.46 px
```

Predicting Pedestrian Trajectory via LSTMs



Demo Time!

Model Outputs



Model Outputs



Ablation Studies

- Obtain more data with diverse conditions (occlusions, weather) and ensure consistent video scaling for LSTM future modeling.
- Add validation split and experiment with parameters/hyper-parameters. LSTM sequence length fine tuning....
- Include more input dimensionality (traffic light states)
- Kinematic Analysis via velocity/acceleration interpolation

Next Steps

